

Q3 Let V be the set of ordered pairs of real numbers. If (a_1, a_2) and (b_1, b_2) are elements of V and $c \in \mathbb{R}$, define

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 b_2) \text{ and } c(a_1, a_2) = (ca_1, a_2).$$

Is V a vector space over $\mathbb{R}$ with these operations? Justify your answer.

Solution.

No. Consider scalars $c = 1, d = 2$ and $(a_1, a_2) = (1, 2)$.

But

$$(c + d)(a_1, a_2) = (1 + 2)(1, 2) = 3(1, 2) = (3, 2)$$

$$c(a_1, a_2) + d(a_1, a_2) = 1(1, 2) + 2(1, 2) = (1, 2) + (2, 2) = (3 + 2, 2 \cdot 2) = (5, 4)$$

This implies that $(c + d)(a_1, a_2) \neq c(a_1, a_2) + d(a_1, a_2)$, thus V is not a vector space over $\mathbb{R}$ with the above operations. $\square$